

and 2024. It was over 143 GW at the end of FY24. Favourable policy, falling solar equipment prices and a strong thrust on decarbonisation in the industrial segment have been the key drivers for this growth.

“Going forward, total renewable capacity additions (solar and wind energy) are expected to be robust at 250-270 GW over FY25-29. This growth will be led by utility-scale projects driven by continued favourable economics, aggressive plan to award 50 GW annually (till FY28), ambitious technology-wise RPO targets as well as policy and regulatory support,” Master tells *Business Today*. RPO is renewable purchase obligation.

The rooftop solar market, led by favourable economics and strong government support through the PM Surya Ghar scheme, is also expected to encourage healthy installations. The FY26 Union Budget has allocated ₹20,000 crore for the scheme.

India’s annual fuel import bill is around \$150 billion, and alternative sources of energy can play a big role in reducing it keeping in mind the abundance of solar and cost economics in favour of renewables.

Nikhil Dhingra, CEO of solar energy company ACME Solar Holdings, says renewables will play a major role in years to come, keeping in mind the country’s huge fuel import cost. He says India has not been able to set up renewable capacity because of cost competitiveness. “Both on the commercial and environmental front, renewable will rank as the most optimal form of energy in which India can be competitive and benefit manufacturing, consumers and producers like us,” Dhingra tells *BT*.

ACME has contributed to 5% of India’s total solar capacity of 100 GW. On the changing course

# ROAD MAP FOR 2030

**500 GW of non-fossil fuel power is expected to be generated**

**This will require at least 336 GWh energy storage capacity**

**₹30-32 lakh crore funding needed to achieve the goal**

**3 million jobs expected to be created in green energy sector**

**Over 8,00,000 acres of land will be needed to set up the projects**



of renewable power, Dhingra says the industry is working to solve the peak demand problem to compete with round-the-clock power in non-renewable hours.

“The adoption of renewables is set to increase very fast in the next 5-10 years as we are solving the peak demand problem. We have been doing hybrid power, battery-based contracts and firm and dispatchable renewable power,” he adds.

## JOURNEY OF TRANSITIONING TO GREEN SOURCES

A recent report by FICCI titled ‘Powering India’s Energy Transition’ points out that meeting India’s target of 500 GW non-fossil fuel capacity by 2030 will require more than incremental changes. The complexity of integrating large-scale renewable energy, ensuring grid reliability, and keeping costs manageable requires a comprehensive approach across three critical pillars: policy, infrastructure, and financing.

On the policy front, India has already laid a strong foundation with its comprehensive policy frameworks, which have driven growth for renewables. The report says the country has seen substantial successes through policy initiatives to address demand-side and supply-side issues, and in overall ecosystem development.

When it comes to infrastructure, the pace of deployment is increasingly constrained by land acquisition challenges and transmission infrastructure bottlenecks. Streamlined land acquisition processes and proactive evacuation planning will be crucial, modelled after successes in states such as Gujarat and Karnataka.

Thirdly, in financing, the money required to meet India’s energy